

Project Report Format

1. INTRODUCTION

1.1 Project Overview

Implement Client Script & UI Policy (Incident) is a ServiceNow micro project that shows how client-side controls enforce data integrity on Incident records. A UI Policy named *High Impact Control* runs when Impact is 1 - High: it makes Assignment group mandatory and, through a second UI Policy Action, makes Urgency read-only. With *Reverse if false* enabled, these effects are undone when Impact is changed to another value.

Three Client Scripts add behaviour: an **onChange** script sets Urgency to High automatically when Impact changes to High, an **onSubmit** script stops a High impact incident from being saved without Assigned to, and an **onCellEdit** script blocks State changes made directly in the list view. The whole configuration was built and tested on the Incident table.

1.2 Purpose

Incident triage, routing and resolution depend on complete and consistent data. Relying on user awareness and manual checks leads to incomplete or incorrect records, which hurts reporting accuracy, SLA compliance and service quality. The purpose of this project is to enforce conditional field behaviour and save-time validation directly on the form, so Incidents meet the defined business rules before they are submitted.

2. IDEATION PHASE

2.1 Problem Statement

Incident records often require consistent and accurate data entry, but manual checks allow incomplete, inconsistent or incorrect data to be submitted. Two customer problem statements were defined:

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a service desk agent who creates and triages Incident records	log high-impact incidents with complete data (Assignment group, Assigned to, Urgency)	records are saved with missing or inconsistent data	the form has no enforced conditional rules and relies on manual checks	frustrated and worried about SLA breaches and wrongly routed tickets
PS-2	a ServiceNow administrator responsible for Incident data quality	enforce rules so every Incident is valid before it is saved or changed	users can bypass checks, e.g. by editing State in the list view	no client-side controls (UI Policy and Client Scripts) validate fields, saves and list edits	concerned about inaccurate reports and poor service quality

2.2 Empathy Map Canvas

Persona: the service desk agent from PS-1, who must log and route high-impact incidents quickly.

SAYS	THINKS
"I need to log this critical incident now." "Which group should this go to?" "Who is this assigned to?"	"Did I fill in every required field?" "Will this breach the SLA?" "Is the urgency consistent with the impact?"
DOES	FEELS
Opens the Incident form, sets Impact and Urgency, leaves Assigned to or Assignment group empty, saves, and relies on manual checks. Edits State directly in the list.	Rushed, worried about misrouted tickets and SLA breaches, frustrated by rework and corrections.
PAINS	GAINS
Missing assignment details; inconsistent Urgency; invalid changes slipping in through list edits.	Guided form: mandatory and read-only fields, automatic Urgency, blocked invalid saves, clean and consistent data.

2.3 Brainstorming

Twelve ideas were generated, grouped into four themes (field behaviour control, automation and guidance, validation and blocking invalid saves, governance and backup checks) and prioritised on importance versus feasibility. Ideas 1 to 7 and 12 were implemented in this project; ideas 8 to 11 are future scope.

#	Idea	Priority	Status
1	Assignment group mandatory when Impact = High (UI Policy)	High	Implemented
2	Urgency read-only when Impact = High (UI Policy action)	High	Implemented
3	Auto-set Urgency to High when Impact changes to High (onChange)	High	Implemented
4	Block save if Assigned to is empty for High impact (onSubmit)	High	Implemented
5	Block State edits from the list view (onCellEdit)	High	Implemented
6	'Reverse if false' so fields revert when Impact is not High	High	Implemented
7	Info message when Urgency is auto-set	Medium	Implemented
8	Server-side Business Rule as backup validation	Medium	Future scope
9	Data Policy for imports and API-created incidents	Low	Future scope
10	Dashboard to monitor incidents with missing data	Low	Future scope
11	Train agents on mandatory-field rules	Low	Future scope
12	Inline error box on the field that fails validation	Medium	Implemented

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Stage	User action	Pain point (before)	Control applied	Experience (after)
Create	Opens Incident > Create New	Nothing guides which fields matter	UI Policy High Impact Control ready on the form	Form behaves according to Impact
Set Impact	Selects Impact = 1 - High	Assignment group can be skipped	Assignment group becomes mandatory	Required field is clearly flagged
Urgency	Reviews Urgency	Urgency inconsistent with Impact	onChange sets Urgency = High; field read-only	Consistent value, no manual step
Submit	Clicks Submit without Assigned to	Incomplete record is saved	onSubmit script blocks save and shows an error box	Clear message: Assigned To is mandatory
Correct	Selects Assigned to and submits	Rework after saving	Record passes validation	Incident saved successfully
Change / list	Lowers Impact; edits State in list	Rules stay locked; State bypass	Reverse if false; onCellEdit alert	Fields revert; State changed only via form

3.2 Solution Requirement

ID	Type	Requirement
FR-1	Functional	When Impact is 1 - High, Assignment group must be mandatory (UI Policy action).
FR-2	Functional	When Impact is 1 - High, Urgency must be read-only (UI Policy action).
FR-3	Functional	When Impact changes to High, Urgency is set to 1 - High automatically (onChange).
FR-4	Functional	A High impact Incident cannot be saved without Assigned to; an error is shown on the field (onSubmit).
FR-5	Functional	State cannot be changed from the list view; an alert is shown and the value is unchanged (onCellEdit).
FR-6	Functional	When Impact is no longer High, policy effects are reversed (Reverse if false = true).
NFR-1	Usability	Messages and field states must make the required action obvious to the user.
NFR-2	Performance	Controls run client-side and stay lightweight, with no extra server round trips.
NFR-3	Maintainability	Configuration uses standard UI Policies and short Client Scripts, easy to adjust.
NFR-4	Compatibility	Applies to the Incident table, form and list, with the Desktop UI type.

3.3 Data Flow Diagram

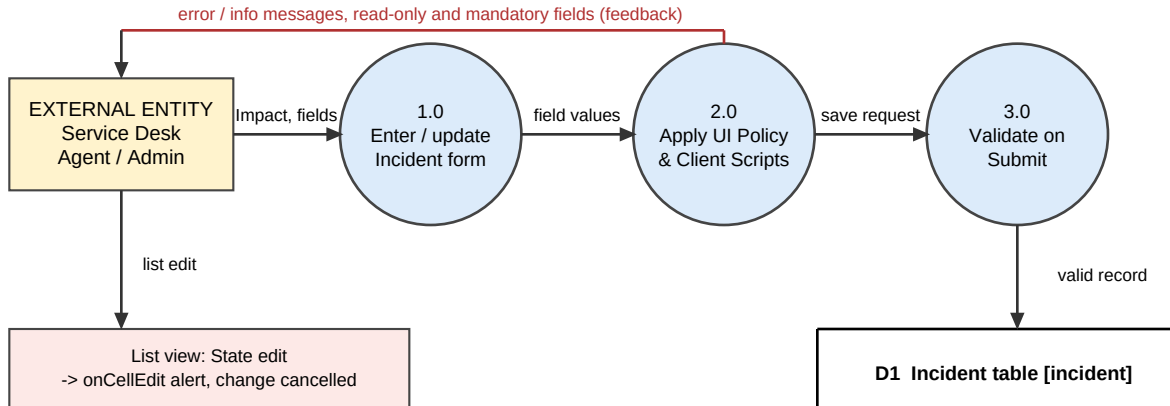


Figure 1: Data flow for the Incident form and list view.

3.4 Technology Stack

Layer	Technology / Component
Platform	ServiceNow Now Platform (instance dev354432.service-now.com)
Module	Incident Management (table: Incident [incident])
Declarative controls	UI Policy and UI Policy Actions (System UI > UI Policies)
Scripting	Client Scripts in JavaScript using g_form (onChange, onSubmit, onCellEdit)
Client	Web browser, Desktop UI type, isolated client scripts

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Solution
Incomplete records for high-impact incidents	UI Policy makes Assignment group mandatory; onSubmit blocks save without Assigned to
Inconsistent Urgency versus Impact	onChange auto-sets Urgency; UI Policy action makes it read-only
Rules left in place after Impact changes	Reverse if false = true reverts the policy effects
Bypass through list editing	onCellEdit alerts the user and cancels the State change

4.2 Proposed Solution

Enforce the rules at the user interface with standard ServiceNow client-side features instead of relying on user awareness. The configuration consists of:

Component	Type / Table	Condition or field	Behaviour
High Impact Control	UI Policy / Incident	Impact is 1 - High; Reverse if false = true	Applies the two actions below
Action: Assignment group	UI Policy Action	Assignment group	Mandatory = True; Read only = False
Action: Urgency	UI Policy Action	Urgency	Read only = True; Visible = Leave alone

Component	Type / Table	Condition or field	Behaviour
Auto set urgency for high impact	Client Script / onChange	Impact	If new value is 1, set urgency to 1 and show info message
Prevent save if Assigned To missing	Client Script / onSubmit	impact, assigned_to	If impact is 1 and assigned_to empty: error box, return false
Prevent state change via list edit	Client Script / onCellEdit	State	Alert and callback(false)

4.3 Solution Architecture

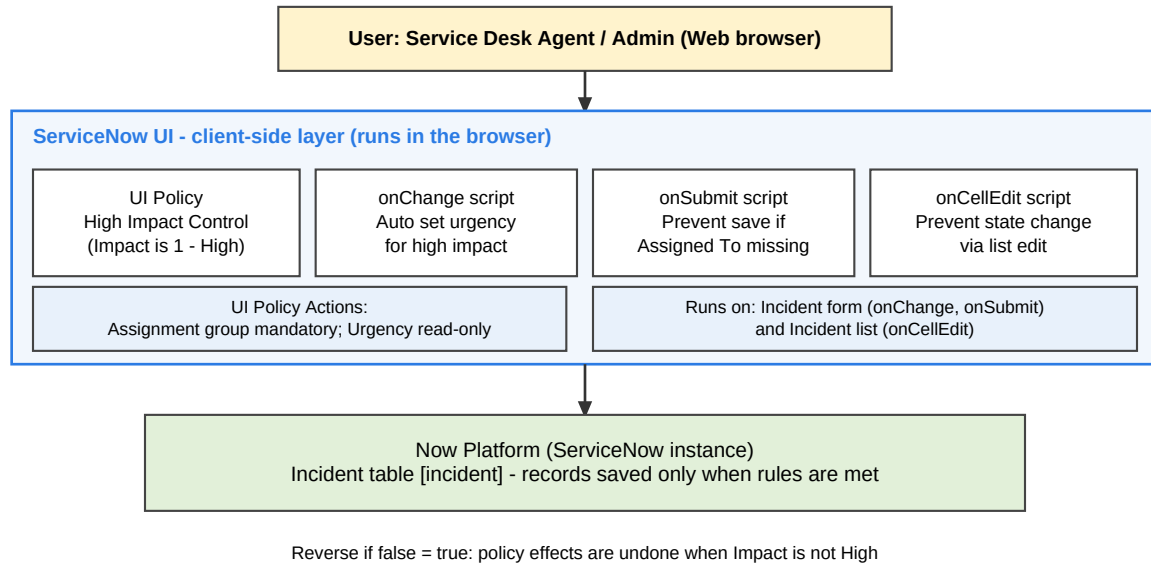


Figure 2: Client-side control layers on the Incident form and list.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Task	Activity	Deliverable	Status
Task 1	Create UI Policy on Incident (High Impact Control, Assignment group mandatory)	UI Policy and action	Completed
Task 2	Create UI Policy Action for Urgency (read-only)	UI Policy action	Completed
Task 3	Create onChange Client Script	Auto set urgency script	Completed
Task 4	Create onSubmit Client Script (save validation)	Prevent save script	Completed
Task 5	Create onCellEdit Client Script	List edit blocking script	Completed
Task 6	Test configuration: enforcement, successful save, reverse condition, list edit blocking, form update	Test results and screenshots	Completed (14 Jan 2026)

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The configuration was tested manually on the Incident form and list with positive and negative scenarios. All 10 test cases passed, with 0 failed and 0 not tested. The controls run in the browser, so they add no server-side processing.

ID	Test scenario	Expected result	Result
TC-001	Assignment group mandatory when Impact = High (UI Policy)	Assignment group becomes mandatory	Pass
TC-002	Urgency read-only when Impact = High (UI Policy Action)	Urgency is read-only	Pass
TC-003	Urgency auto-set for High impact (onChange)	Urgency set to 1 - High automatically	Pass
TC-004	Negative: save without Assigned to (onSubmit)	Save blocked; error shown on Assigned to	Pass
TC-005	Positive: save with Assigned to filled (onSubmit)	Incident saved without errors	Pass
TC-006	State edit blocked in list view (onCellEdit)	Alert shown; State unchanged	Pass
TC-007	Reverse if false: Assigned to not mandatory	Assigned to is not mandatory	Pass
TC-008	Reverse if false: Urgency becomes editable	Urgency is editable again	Pass
TC-009	Record saves after Impact changed to Medium	Record saved with no errors	Pass
TC-010	State change from the Incident form	State change saved; other rules still apply	Pass

One observation was logged (by design, low severity): after Impact is changed from High to Medium, Urgency keeps its value of 1 - High because the onChange script only sets Urgency when Impact is High; Urgency is editable again and can be adjusted manually.

7. RESULTS

7.1 Output Screenshots

The screenshot displays the ServiceNow 'Incident - Create INC0010012' form. The form is partially filled with test data. A red box highlights the 'Impact' dropdown menu, which is currently set to '1 - High'. Another red box highlights the 'Assigned to' field, which is empty. A red error message is visible below the 'Assigned to' field, stating: 'Assigned To is mandatory for High impact incidents.' The 'Submit' button is also highlighted with a red box. The left sidebar shows the 'Incident' menu with options like 'Create New', 'Assigned to me', 'Open', 'Open - Unassigned', 'Resolved', 'All', 'Overview', 'Critical Incidents: Map', 'Administration', and 'Incident Properties'. The bottom of the form shows a 'Related Search' section with a search bar containing 'testing UI policy' and a 'Knowledge & Catalog (AI)' dropdown.

Figure 3: Incident not saved; error shown because Assigned To is empty for High impact.

servicenow All

Incident - Create INC0010012

Incident New record

Number: INC0010012

* Caller: Adela Cervantsz

Category: Inquiry / Help

Subcategory: -- None --

Service:

Service offering:

Configuration item:

* Short description: testing UI policy

Description:

Channel: -- None --

State: New

Impact: 1 - High

Urgency: 1 - High

Priority: 1 - Critical

* Assignment group: Software

Assigned to: Beth Anglin

Submit Resolve

Related Search Results

Related Search: Q testing UI policy Knowledge & Catalog (AI)

Company policies No description

Whitelist IP Need to whitelist an IP address? Order

Firewall Rule Change Cisco Firewall Appliance Order

Figure 4: Incident saved after Assigned to is filled; Urgency auto-set and read-only.

servicenow All

Incident - INC0010012

Incident INC0010012

Number: INC0010012

* Caller: Adela Cervantsz

Category: Inquiry / Help

Subcategory: -- None --

Service:

Service offering:

Configuration item:

* Short description: testing UI policy

Description:

Channel: -- None --

State: New

Impact: 2 - Medium

Urgency: 1 - High

Priority: 2 - High

Assignment group: Software

Assigned to:

Discuss Follow Update Resolve Delete

Related Search Results

Country: -- None --

States: -- None --

Notes Related Records Resolution Information

Figure 5: Reverse condition; Impact changed to Medium and Assigned to no longer mandatory.

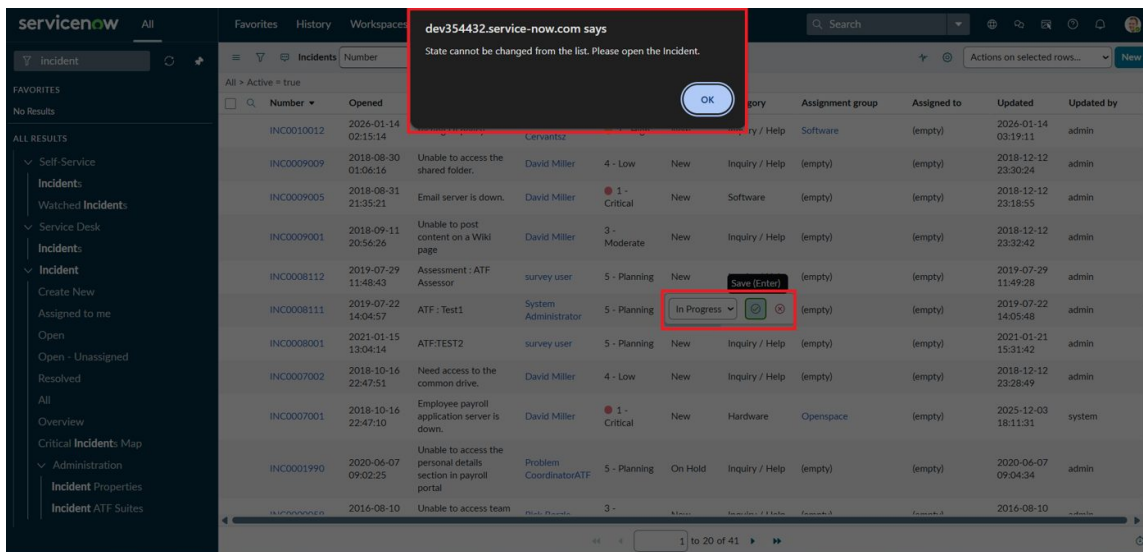


Figure 6: Alert shown when State is edited from the list view.

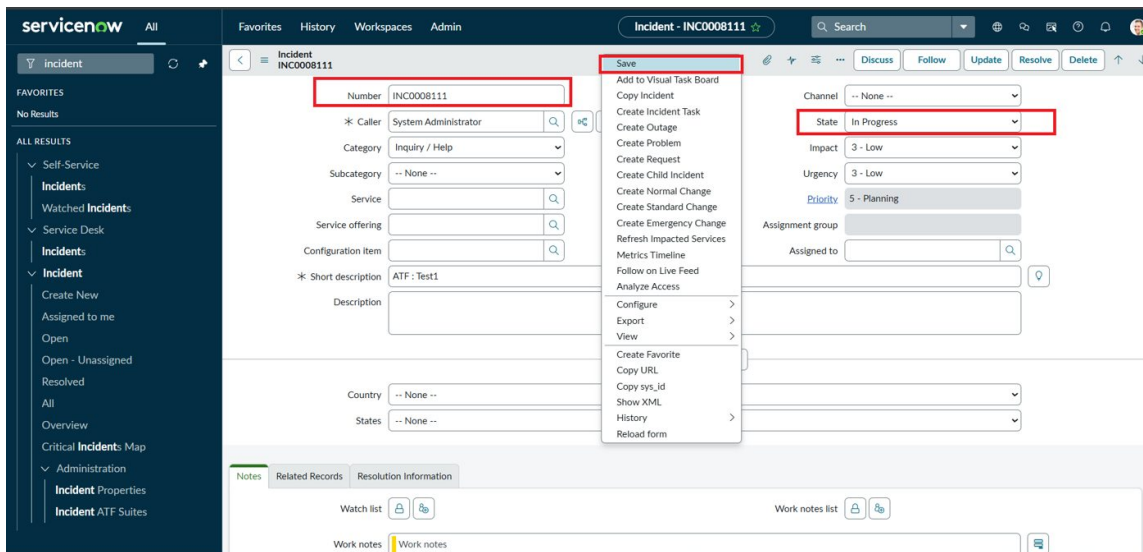


Figure 7: State changed on the Incident form (allowed).

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INCO0010012	2026-01-14 02:15:14	testing UI policy	Adela Cervantsz	2 - High	New	Inquiry / Help	Software	(empty)	2026-01-14 03:19:11	admin
INCO009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INCO009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INCO009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INCO008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin
INCO008111	2019-07-22 14:04:57	ATF : Test1	System Administrator	5 - Planning	In Progress	Inquiry / Help	(empty)	(empty)	2026-01-14 03:19:11	admin
INCO008001	2021-01-15 13:04:14	ATF:TEST2	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2021-01-21 15:31:42	admin
INCO007002	2018-10-16 22:47:51	Need access to the common drive.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:28:49	admin
INCO007001	2018-10-16 22:47:10	Employee payroll application server is down.	David Miller	1 - Critical	New	Hardware	Openspace	(empty)	2025-12-03 18:11:31	system
INCO001990	2020-06-07 09:02:25	Unable to access the personal details section in payroll portal	Problem CoordinatorATF	5 - Planning	On Hold	Inquiry / Help	(empty)	(empty)	2020-06-07 09:04:34	admin
INCO000000	2016-08-10	Unable to access team	Work Processor	3 -	New	Personal / Help	Personal-3	(empty)	2016-08-10	admin

Figure 8: Incident list after the form update; State shows In Progress.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Improves data quality and consistency on Incident records	Client-side controls do not run for API calls, imports or other server-side updates
Immediate feedback to the user before saving	Urgency stays High when Impact is lowered (by design)
Automatic Urgency reduces manual effort	onCellEdit blocks State list edits for every user
Lightweight, quick to build and easy to maintain	Values such as '1' are hard-coded in scripts
Supports reporting accuracy and SLA compliance	Needs regression testing after platform upgrades

9. CONCLUSION

The project shows how UI Policies and Client Scripts work together to enforce dynamic field behaviour, automate updates and prevent incorrect data submission on Incident forms. The onChange, onSubmit and onCellEdit scripts, together with the UI Policy actions, make fields such as Assigned to and Urgency respond to changes in Impact and keep data clean and consistent. The solution is lightweight, efficient and easy to implement, which suits a short-duration micro project and demonstrates good practice for form usability and data integrity in ServiceNow.

10. FUTURE SCOPE

- Add a server-side Business Rule as a backup so the rules also apply to API, import and integration updates.
- Apply a Data Policy to cover records created outside the form.
- Build a dashboard to monitor incidents with missing or inconsistent data.
- Reset or review Urgency automatically when Impact is lowered.
- Replace hard-coded values with system properties or choice values, and add role-based exceptions for list edits.
- Provide agent training and guidelines on the mandatory-field rules.

11. APPENDIX

Source Code(if any)

onChange - Auto set urgency for high impact (Incident, field: Impact)

```
function onChange(control, oldValue, newValue, isLoading) {  
    if (isLoading || newValue == '') {  
        return;  
    }  
    if (newValue == '1') {  
        g_form.setValue('urgency', '1');  
        g_form.addInfoMessage('Urgency set to High for High impact incident.');    }  
}
```

onSubmit - Prevent save if Assigned To missing (Incident)

```
function onSubmit() {  
    if (g_form.getValue('impact') == '1' &&  
        g_form.getValue('assigned_to') == '') {  
        g_form.showErrorMessage(  
            'assigned_to',  
            'Assigned To is mandatory for High impact incidents.'  
        );  
        return false;  
    }  
    return true;  
}
```

onCellEdit - Prevent state change via list edit (Incident, field: State)

```
function onCellEdit(sysIDs, table, oldValues, newValue, callback) {  
    alert('State cannot be updated using list editing. Please open the Incident.');    callback(false);  
}
```

Dataset Link

Not applicable. The project uses the platform's Incident table; no external dataset was used.

GitHub & Project Demo Link

Not provided (no repository or demo link was supplied with the project).